

altogether another distribution that is being inferred, that of the best performers. We call the difference between the average of such distribution and the unconditional distribution of winners and losers the *survivorship bias* – here the fact that about 3% of the initial cohort will make money five years in a row. In addition, this example illustrates the properties of *ergodicity*, namely, that time will eliminate the annoying effects of randomness. Looking forward, in spite of the fact that these managers were profitable in the past five years, we expect them to break even in any future time period. They will fare no better than those of the initial cohort who failed earlier in the exercise. Ah, the long term.

A few years ago, when I told one A., a then Master-of-the-Universe type, that track records were less relevant than he thought, he found the remark so offensive that he violently flung his cigarette lighter in my direction. The episode taught me a lot. Remember that nobody accepts randomness in his own success, only his failure. His ego was pumped up as he was heading up a department of “great traders” who were then temporarily making a fortune in the markets. They subsequently blew up during the harsh New York winter of 1994 (it was the bond market crash that followed the surprise interest rate hike by Alan Greenspan). The interesting part is that six years later I can hardly find any of them still trading (ergodicity).

Recall that the survivorship bias depends on the size of the initial population. The information that a person made money in the past, just by itself, is neither meaningful nor relevant. We need to know the size of the population from which he came. In other words, without knowing how many managers out there have tried and failed, we will not be able to assess the validity of the track record. If the initial population includes ten managers, then I would give the performer half my savings without a blink. If the initial population is composed of 10,000 managers, I would ignore the results. The latter situation is generally the case; these days so many people have been drawn to the financial markets. Many college graduates are trading as a first career, failing, then going to dental school.

If, as in a fairy tale, these fictional managers materialized into real human beings, one of these could be the person I am meeting tomorrow at 11.45 a.m. Why did I select 11.45 a.m.? Because I will question him

about his trading style. I need to know how he trades. I will then be able to claim that I have to rush to a lunch appointment if the manager puts too much emphasis on his track record.

Life Is Coincidental

Next we look at the extensions to real life of our bias in the understanding of the distribution of coincidences.

THE MYSTERIOUS LETTER

You get an anonymous letter on January 2nd informing you that the market will go up during the month. It proves to be true, but you disregard it owing to the well-known January effect (stocks have gone up historically during January). Then you receive another one on Feb 1st telling you that the market will go down. Again, it proves to be true. Then you get another letter on March 1st – same story. By July you are intrigued by the prescience of the anonymous person and you are asked to invest in a special offshore fund. You pour all your savings into it. Two months later, your money is gone. You go spill your tears on your neighbor's shoulder and he tells you that he remembers that he received two such mysterious letters. But the mailings stopped at the second letter. He recalls that the first one was correct in its prediction, the other incorrect.

What happened? The trick is as follows. The con operator pulls 10,000 names out of a phone book. He mails a bullish letter to one half of the sample, and a bearish one to the other half. The following month he selects the names of the persons to whom he mailed the letter whose prediction turned out to be right, that is, 5,000 names. The next month he does the same with the remaining 2,500 names, until the list narrows down to 500 people. Of these there will be 200 victims. An investment in a few thousand dollars worth of postage stamps will turn into several million.